

FINANCIAL DASHBOARD ANALYSIS REPORT



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COURSE: Data Visualization in Economics
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Agenda

- Introduction
- Data Processing & Modeling
- Visualiztion
- Analysis insight
- Conclusion





INTRODUCTION

Project objective:

- To develop an **automated financial analysis dashboard** for Vietnam International Commercial Joint Stock Bank (VIB) on the Power BI platform.
- The system is designed to process, standardize, and visualize data from two critical financial sources: the **Income Statement** and the **Balance Sheet**.

Core Challenges:

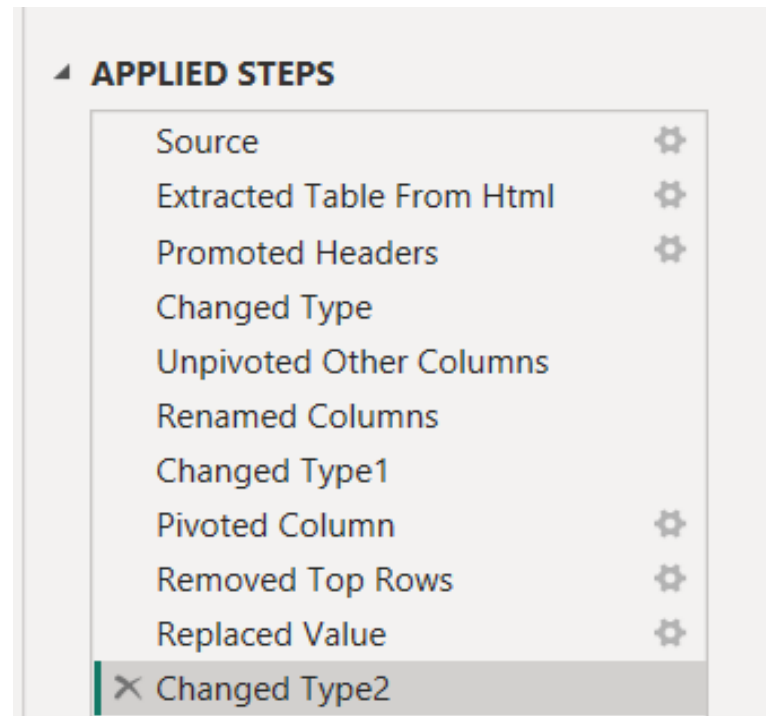
- The primary technical hurdle involves resolving a significant **timeframe discrepancy**: the Income Statement data is updated through 2025, whereas the Balance Sheet data currently concludes in 2023. To address this, the project implements **advanced data modeling** to ensure accurate cross-report synchronization and reliable year-over-year (YoY) metric comparisons.



Data Processing & Modeling

Data Transformation (Power Query):

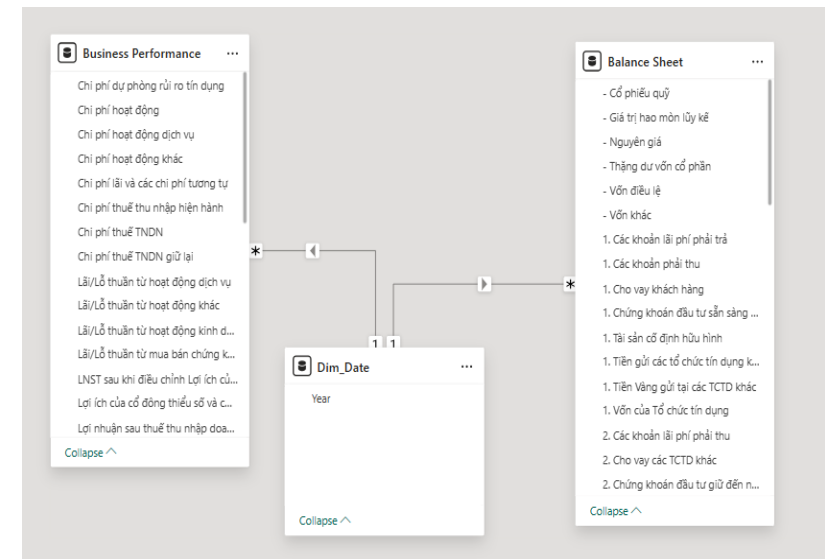
- **Unpivot Technique:** Converted data from a wide format (multiple year columns) into a **long format (tabular)**. This transformation optimizes storage efficiency and is essential for developing dynamic DAX measures and trend visualizations like Sparklines.
- **Data Cleaning:** Removed redundant prefixes (e.g., "Year "), and standardized data types to **Whole Number** for years and **Decimal Number** for financial metrics to ensure calculation accuracy.



Advanced Modeling:

- **Dim_Date:** Created a centralized calendar table (covering 2019–2025) to act as a unified timeline for both the Income Statement and Balance Sheet.
- **Schema Architecture:** Established **One-to-Many (1:*)** relationships from the Dim_Date table to the primary datasets. This architecture successfully resolves the timeframe discrepancy (2023 vs. 2025), ensuring that all slicers and visuals remain synchronized across the entire dashboard..

```
1 Dim_Date =  
2 SELECTCOLUMNS(  
3     GENERATESERIES(2019, 2025, 1),  
4     "Year", [Value]  
5 )
```





KEY DAX MEASURES FOR FINANCIAL ANALYSIS

1. Profit & Loss (P&L) Indicators:

- These measures reflect VIB's core operational efficiency and profitability. All values are scaled to **Billion VND** (converted from Million VND) for professional reporting.

```
1 Gross Profit = [Total Revenue] - [COGS]
```

```
1 Total Revenue = DIVIDE(SUM('Business Performance'[Thu nhập lãi thuần]) + SUM('Business Performance'[Lãi/Lỗ thuần từ hoạt động dịch vụ]) + SUM('Business Performance'[Lãi/Lỗ thuần từ hoạt động khác]), 1000)
```

```
1 COGS = DIVIDE(ABS(SUM('Business Performance'[Chi phí lãi và các chi phí tương tự])), 1000)
```

```
1 Operating Expenses = DIVIDE(ABS(SUM('Business Performance'[Chi phí hoạt động])), 1000)
```

2. Balance Sheet Aggregates:

- These measures automatically responds to Year/Quarter slicers via Dim_Date. This also returns **BLANK** for 2024-2025 (instead of 0) and : prevents "Data Hallucination"—clearly signals to users that Balance Sheet data is pending, while P&L remains visible.

```
1 Customer Deposits = DIVIDE(SUM('Balance Sheet'[III. Tiền gửi khách hàng]),1000)
```

```
1 Total Assets = DIVIDE(SUM('Balance Sheet'[TỔNG CỘNG TÀI SẢN]),1000)
```

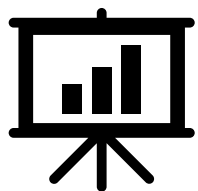
3. Financial Ratios & Growth Analytics:

- Uses **DIVIDE** to prevent **#DIV/0!** errors when data is missing.
- **Time Intelligence:** Uses **VAR** (Variables) to capture the current selection and **ALL('Dim_Date')** to "break" the slicer's filter.
- **Prior Year Retrieval:** The **CALCULATE** function allows the model to "reach back" to the previous year (**\$CurrentYear - 1\$**) to fetch the comparative baseline.
- **Safe Calculation:** Employs the **DIVIDE** function to handle years with no data, ensuring the Dashboard remains error-free (avoiding **#DIV/0!**).

```
1 NIM = DIVIDE('Business Performance'[Revenue], [Total Assets])
```

```
1 Growth % Revenue =  
2 VAR CurrentYear = SELECTEDVALUE('Dim_Date'[Year])  
3 VAR CurrentRev = [Total Revenue]  
4 VAR PriorRev =  
5     CALCULATE(  
6         [Total Revenue],  
7         ALL('Dim_Date'),  
8         'Dim_Date'[Year] = CurrentYear - 1  
9     )  
10 RETURN  
11     DIVIDE(CurrentRev - PriorRev, PriorRev)
```

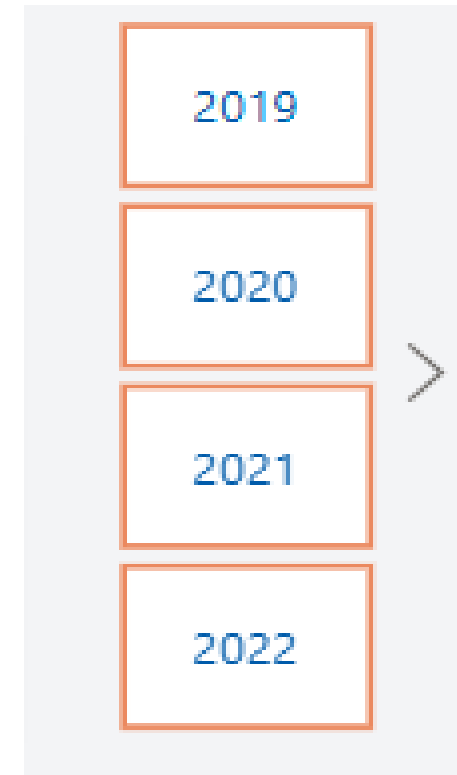
VISUALIZATION & UI/UX STRATEGY



Vertical Tile UI: Quick navigation and clear visibility of the selected reporting period.

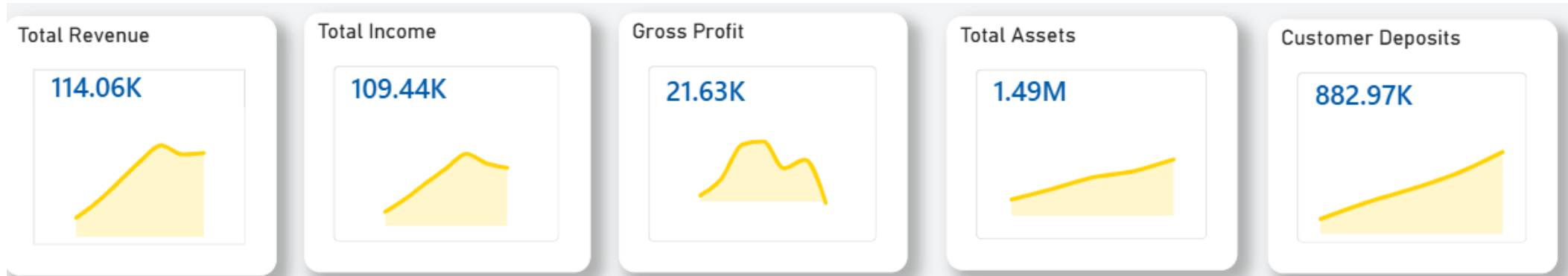
Unified Dim_Date Engine: A single bridge table controlling multiple datasets → One-click synchronization for both P&L (2025) and Balance Sheet (2023).

Smart Filter Logic: Handles the timeframe discrepancy automatically → Selecting 2019–2025 updates Income visuals while keeping Balance Sheet cards empty, ensuring **Data Integrity**.



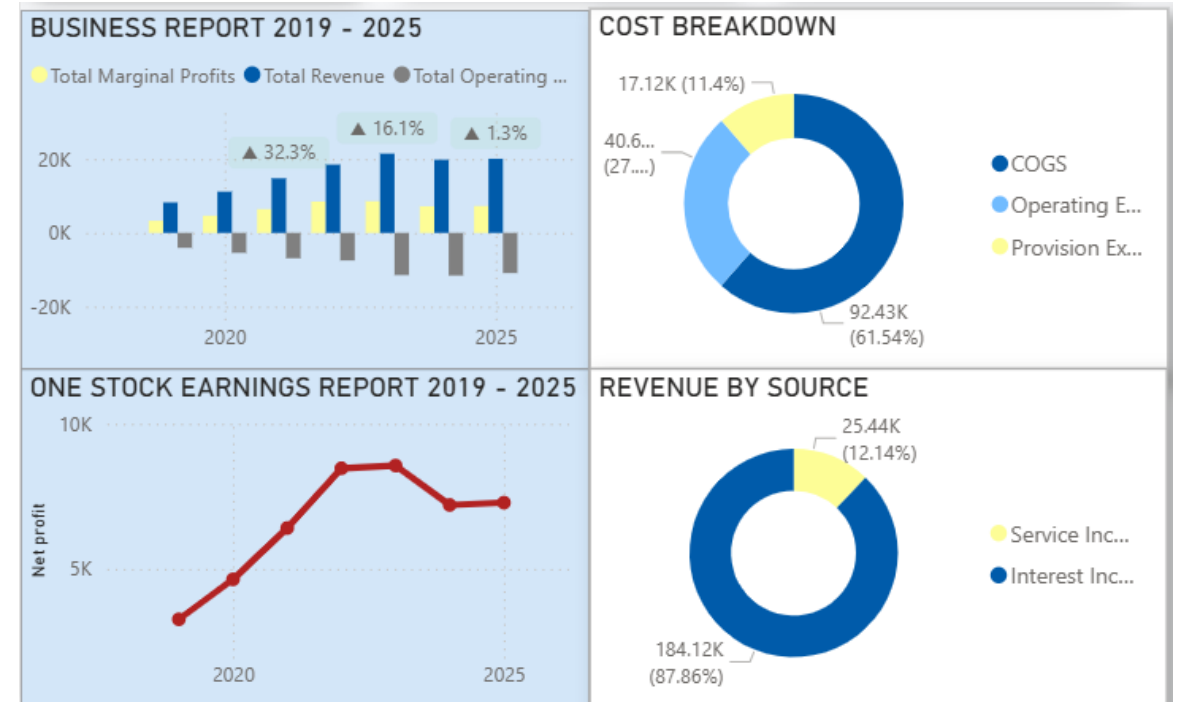
Interactive slicer – Central control

- **Integrated KPI Cards (Current Status):** Displays core metrics (Revenue, Profit, Assets) in high-contrast blocks → immediately upon opening the dashboard.
- **Embedded Sparklines (Yellow Trend Lines):** Layering 7-year historical data directly behind current numbers → Provides **Instant Context**. Users see the bank's long-term momentum (Growth/Dips) without needing to hover, click, or adjust slicers.
- **"Zero-Click" Navigation Design:** Minimalist layout that removes axes, grids, and "Chart Junk." → Forces focus on the **Trajectory** rather than raw volume, allowing the eye to scan the bank's total health in seconds.



High-Impact KPI Hierarchy (New Card Visual)

- **Clustered Column Charts:** Side-by-side Revenue vs. Profit. → **Floating % Labels** (Dynamic Delta). Instant YoY growth tracking without reading axes.
- **Donut Charts :**80% Inner Radius (Modern UI) → **Cost & Revenue Mix** (COGS vs. OPEX). Identifies Concentration Risk & Portfolio Diversification.
- **Line Charts:** Continuous timeline tracking. → **High-contrast Markers**. Visualizing long-term stability vs. short-term fluctuations.



Charts (Performance Benchmarking)

ANALYSIS INSIGHTS



STRATEGIC SUMMARY



1. P&L Performance Momentum (2019 – 2025):

- TOI, Net Interest Income, and NPAT trajectory.
- Correlating Revenue growth vs. OPEX efficiency to monitor core profitability.

2. Balance Sheet Composition (Up to 2023):

- Funding Mix (Deposits vs. Debt Issuance) and Asset Allocation.
- Analyzing structural shifts in capital and liability management.

3. Core Banking Ratios & Health:

- Monitoring NIM (Net Interest Margin) and ROE trends.
- Evaluating operational efficiency and shareholder value creation.

4. Timeframe Synchronization (Addressing the Data Gap):

- Synchronized analysis between 2025 P&L results and 2023 Balance Sheet snapshots.
- Ensures Year-over-Year (YoY) accuracy while acknowledging data availability limits.

Comparative Performance Summary

Metric	2023 Performance	2024 Performance	Status
Total Assets (Billion)	409,881	493,158	+20% Growth
Total Revenue (Billion)	22,160	20,570	-7% Decline
Customer Deposits (Billion)	236,577	276,308	+17% Growth
Pre-tax Profit (Billion)	10,700	9,000	-16% Decline

Industry Benchmarking & Trends

VIB Credit Growth

22%

Market Avg Growth

14%

VIB NIM

3.8%



VIB demonstrates industry-leading credit growth (22%) despite external market pressures, successfully leveraging its retail-focused strategy and digital transformation initiatives.



Thank you

VIB DASHBOARD ANALYSIS